Memory Stick Controller Design Specification

Ver 0.0

DM270

Department: IMAGING

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HISTORY

Version	Date	Author	Notes
Ver 0.0	05/14/02	Kazuhiro Yano	Initial Release for DM270

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1. ABSTRACT

本メモリースティックコントローラーは16bit CPUインターフェイスを持つ“Memory Stick Ver. 1.3”及び“Memory Stick Pro”対応ホストコントローラーです。 本モジュールは“Memory Stick Standard Fromat Specification ver. 1.3“及び”Memory Stick Pro Format仕様書 Ver. 0.8(J)に準拠します。

2. REFERENCES

Memory Stick Standard Format Specification ver. 1.3.
Memory Stick Pro Fromat Specification ver. 0.8(J).

3. FEATURE

- データ送受信用FIFO（64bit X 4）を内臓。
- CRC回路内臓。
- メモリースティックシリアルクロック（Ver.1.3 : Max 20Mhz, Pro : Max 40Mhz）。

4. FUNCTION

4.1

CPUからCommand Registerへの書き込みによりメモリースティックとの
通信プロトコルを開始します。プロトコル終了時は、CPUに対し割り込み要求により
プロトコルの終了を通知します。

4.2

メモリースティックとの通信でハンドシェイク状態（リードプロトコル時：BS2、ライトプロトコル時：BS3）の
際のRDYタイムアウト時間をメモリースティック転送クロック数で指定することが出来ます。
タイムアウトした場合はCPUに対して割り込み要求によりタイムアウトエラーによるプロトコル終了を
通知します。

4.3 CRC

テストモードとしてCRCオフの設定が可能です。 但しCRCオフの設定時は
メモリースティックへのデータ転送はCRCを付加しません。

4.4

プロトコルが開始されデータ転送期間に入るとCPU割り込みによる
データ転送要求も可能です。

4.5

System　RegisterのINTENが“1”の時に割り込み要求が許可されます。
Status　RegisterのDRQ，MSINT、 MSRDYが割り込み要因となり、各ビットが
“0”から“1”へ変化した際に割り込み要求が発生します。 但しMSINTは他の割り込み
要因による割り込みが発生している場合は、割り込みをクリアした後にMSINTによる
割り込みが発生します。 割り込みはSystem　RegisterのMSINTCLRを“1”にすることで
クリアできます。 但しSystem　RegisterのMSDRQSLに“1”が設定されている場合は
データ転送要求（MSDRQ）による割り込みが発生した時FIFOへの書き込み、または
読み出しでも割り込み要求のクリアは可能です。

5. Register

レジスタ空間 : 0x0003:0C80~0x0003:0CFF

5.1 Mode Register

Mode Register（0003:0C80） OFFSET＝0x0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	MODE
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved Bits. Do not use.
0	MODE	Memory Stick Interface Enable Bit. 0 : MMC Interface is enable. 1 : Memory Stick Interface is enable.

5.2 Command Register

Command Register（0003:0C82） OFFSET＝0x1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	TPC3	TPC2	TPC1	TPC0	RSV	RSV	DSZ9	DSZ8	DSZ7	DSZ6	DSZ5	DSZ4	DSZ3	DSZ2	DSZ1	DSZ0
R/W	R/W	R/W	R/W	R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/\w	R/W	R/W
Default	0	0	0	0	-	-	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:12	TPC3-TPC0(*)	Transfer Protocol Code. These bits designate the TPC.
11:10	RSV	Reserved Bits. Do not use.
9:0	DSZ9-DSZ0(*)	Data Size Bits. These bits designate the transmit/receive data lentgh. The length can be set from 1byte to 1024bytes. When DSZ is 0, data size is set to 1024bytes.

(*)Note: 1. When the command register is written, the communication protocol with the Memory Stick starts and data transmit/receive is performed. The data transfer direction with the Memory Stick is determined from TPC3. When TPC3 is 0, the read protocol is performed, and when TPC3 is 1, the write protocol is performed. If the TPC transfer direction and system register bit FDIR differ, TPC3 value is reflected to system register bit FDIR when the protocol starts.

2. The command register is initialized when RST(system register) is 1 and hardware reset is asserted.

3. The command register cannot be written during protocol execution(RDY=0(status register)).

5.3 Data Register

Data Register（0003:0C84） OFFSET＝0x2

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MSD15	MSD14	MSD13	MSD12	MSD11	MSD10	MSD09	MSD08	MSD07	MSD06	MSD05	MSD04	MSD03	MSD02	MSD01	MSD00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/\w	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	MSD15-MSD00(*)	FIFO Data Buffer.
11:10	RSV	Reserved Bits. Do not use.
9:0	MSDSZ9-MSDSZ0(*)	Data Size Bits. These bits designate the transmit/receive data lentgh. The length can be set from 1byte to 1024bytes. When MSDSZ is 0, data size is set to 1024bytes.

(*)Note: 1. This register is used to access the interal FIFO.

The minimum FIFO access unit is 8bytes. Even if transfer/receive data size is less than 8bytes, have to read/write 8bytes of data.

The data is transmitted to and received from the MemoryStick MSB first. When the data is less Than 8bytes, the lower bytes are invalis data.

5.4 Status Register

Status Register(0003:0C86) OFFSET＝0X3

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	DRQ	MSINT	RDY	RSV	RSV	CRC	TOE	RSV	RSV	EMP	FUL	CED	ERR	BRQ	CNK
R/W	-	R	R	R	-	-	R	R	-	-	R	R	R	R	R	R
Default	-	0	0	0	-	-	0	0	-	-	1	0	0	0	0	0

Bits	Register Name	Description
15	RSV	Reserved Bits. Do not use.
14	DRQ	DMA Request bit. DM270 doesn't have DMA Function. This bit should be masked by software. 1: DMA Request(This indicates transmit data request.) 0: No DMA Request.
13	MSINT	MemoryStick Interface Interrupt bit. 1: Interrupt request received from MemoryStick. 0: No Interrupt request received from Memory Stick.
12	RDY	Ready bit. 1: Command receive enabled or protocol ended. 0: Command receive disabled.
11:10	RSV	Reserved Bits. Do not use.
9	CRC	CRC Error Bits. 1: CRC error occurred during data receive. 0: No CRC error.
8	TOE	Time Out Error Bit. 1: RDY time out error . 0: No time out error.
7:6	RSV	Reserved Bits. Do not use.
5	EMP	FIFO Empty. 1: FIFO empty 0: FIFO contains data. This bit is cleared to 1 by writing system register bit FCR=1.

4	FUL	FIFO Full. 1: FIFO full. 0: FIFO has empty spaces. This bit is cleared to 0 by writing system register bit FCR=1.
3	CED	MS Command End. 1: Command end is indicated. 0: Command end is NOT indicated. In parallel interface mode, the contents of the CED bit of Memory Stick Status register are reflected to this bit at the same time as MSINT. In serial interface mode, this bit is always 0. This bit is cleared to 0 by writing to the command register.
2	ERR	MS Error. 1: Memory Stick Error is indicated. 0: No Memory Stick errors. In parallel interface mode, the contents of ERR bit of the Memory Stick Status register are reflected to this bit at the same time as MSINT. In serial interface mode, this bit is always 0. This bit is cleared to 0 by writing to the command register.
1	BRQ	MS Data Buffer Request. 1: Memory stick data buffer requst is indicated. 0: No data buffer request. In parallel interface mode, the contents of the BREQ bit of the Memoy Stick status register are reflected to this bit at the same time as MSINT. In serial interface mode, this bit is always 0. This bit is cleared to 0 by writing to the command register.
0	CNK	MS Command No Acknowledge. 1: Command NO Acknowledge is indicated from MS. 0: Command Acknowledge is indicated. In parallel interface mode, the contents of the CMDNK bit of the Memory Stick status register are reflected to this bit at the same time as MSINT. In serial interface mode, this bit is always 0. This bit is cleared to 0 by writing to the command register.

5.5 System Register

System Register(0003:0C88) OFFSET＝0X4

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	MSRST	SRAC	INTEN	NOCRC	INTCLR	MSIEN	FCLR	FDIR	RSV	RSV	DRQSL	REI	RSV	BSY2	BSY1	BSY0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R\W	-	-	R/W	R/W	-	R/W	R/W	R/W
Default	0	1	0	0	0	1	0	0	-	-	0	0	-	1	0	1

Bits	Register Name	Description
15	MSRST	Sync Reset. 1: Sync reset is performed. 0: No sync reset is performed. MSRST is cleared to 0 after the internal reset is completed.
14	SRAC	Serial Access Mode. 1: Serial interface mode. 0: Parallel interface mode. SRAC setting cannot be changed during protocol execution.
13	INTEN	Interrupt Enalbe. 1: Interrupt request output to HOST enalbe. 0: Interrupt request output to HOST disable.
12	NOCRC	No CRC. 1: CRC output off. 0: CRC output on. When NOCRC is set to 1, the write protocol is executed without CRC data. During the read protocol, the CRC check is performed like normal Reagardless of the NOCRC value. Normally should be set to 0.
11	INTCLR	Interrupt Clear. 1: Interrupt to HOST is cleared. 0: No interrupt clear. Interrupt to host is deasserted by setting INTCLR=1. INTCLR is cleared to 0 after interrupt is deasserted.
10	MSIEN	MS Interrupt Enable. 1: Interrupt request from Memory Stick enabled. 0: Interrupt request from Memory Stick disabled. When MSIEN is set to 0, the status register bit MSINT does not changed and interrupt request to host is not asserted.
9	FCLR	FIFO Clear. 1: FIFO clear enable. 0: FIFO clear disable.
8	FDIR	FIFO Direction. 1: FIFO direction is set to (CPU -> FIFO -> MS). 0: FIFO direction is set to (CPU <- FIFO <- MS).
7:6	RSV	Reserved Bits. Do not use.
5	DRQSL	DMA Request Select. DM270 doesn't have a DMA function. Please set 0 to this bit.

4	REI	Rise Edge Input 1: MSDIO is latched at the rising edge of SCLK. 0: MSDIO is latched at the falling edge of SCLK. Initial value is set to 1.
2:0	BSY2 – BSY0	Busy Count. Initial value is set to 5(BSY[2:0]=101). These bits set the RDY time-out time for the Memory Stick. The maximum BSY wait time until the RDY signal is output from the Memory Stick is set to the (BSY setting value x 4 SCLK). However time-out detection is not performed when BSY=0. When performing wakeup for a vertion 1.3 Memory Stick, the RDY signal from the Memory Stick is extended compared to nomal operation, so set BSY=0 and start the protocol. These bits cannot be changed during protocol execution.

5.6 Endian Register

Endian Register(0003:0C8A) OFFSET＝0X5

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	MSEND
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

15:1	RSV	Reserved Bits. Do not use.
0	MSEND	Memory Stick Endian bit. 1: Big Endian. 0: Little Endian. Initial value is set to 0.

5.7 Interrupt Status Register

Interrupt Status Register(0003:0C8C) OFFSET＝0X6

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	MSINTS
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

15:1	RSV	Reserved Bits. Do not use.
0	MSEND	Memory Stick interrupt status register. 1: Interrupt is asserted. 0: No interrupt is asserted.

6. Interface

Table 1 : BUSC Top-level I/O List

Pin Name	Bits	I/O	Fr/To	Description
Functional clocking / common signals				
rst_n	1	I	ext	Power on reset..
clk_arm	1	I	clkc	ARM clock.
clk_msif	1	I	clkc	Divided arm clock.
clk_msclk	1	I	clkc	MS serial clock.
nclk_msclk	1	I	clkc	MS serial clock inverted.
BUSC interface signals				
busc_ncs	1	I	busc	Memory request from busc.
busc_nrw	1	I	busc	Read/Write direction from busc. 0:Read, 1:Write.
busc_ad	6	I	busc	Address from busc.
busc_di	16	I	busc	Data Input from busc.
ms_busc_do	16	O	busc	Data Output to busc.
ms_nwait	1	O	busc	Nwait to busc.
INTC module interface signals				
ms_nint	1	O	intc	Interrupt signal to intc. 0: interrupt is asserted.
Memory Stick interface signals.				
ms_mode	1	O	top	MS mode enalbe. 0: disable, 1:enable
ms_sclk	1	O	MS	MS serial clock.
ms_bs	1	O	MS	Ms bus state signal.
ms_di	4	I	MS	MS data input bus
ms_do	4	O	MS	MS data output bus.
ms_do_oe	1	O	top	MS data output enable. 0: output, 1:input
TEST control signal				
tst_scan	1	I	tst	Scan test enable.

7. TEST Methodology

None.